



Africa Shield AI

API Schema Reference — for the frontend team

Exact live response shapes, verified against the running backend
(not reconstructed from memory or from an older doc)

Prepared 10 August 2026 · Source of truth: `docs/api-contract.md`

What changed since you last checked

You flagged four gaps against what you needed (location, country, lat/lon, risk level, risk score, rainfall, water level, alert message). All four are now closed — additively, nothing existing was renamed or removed.

1. **country** is now a real field on both endpoints — parsed from `location_name` server-side, so you don't need to split the string yourself.
2. **latitude / longitude** are now echoed back in `POST /api/risk-check` 's response (previously only in the request).
3. **rainfall_mm_24h / river_level_m** are now **top-level** on both endpoints — previously you had to reach into `risk_score_breakdown.rainfall_mm_24h` for these. (They're still there too, for backwards compatibility.)
4. `GET /api/regions` now includes an alert message per region (`alert_message_en` , `alert_message_local` , `local_language`) — previously only `POST /api/risk-check` had this, meaning a per-region list view needed an extra call per city.

HOW THIS WAS PRODUCED

Every example in this document was copy-pasted from an actual running call to the backend on 2026-08-10, cross-checked against the route code in `backend/app/routes/` — not reconstructed from an older doc or from memory.

POST /api/risk-check

Score one location's flood risk from rainfall + river level. Real, computed live — not stubbed.

Request body

```
{
  "location_name": "Lagos, Nigeria",
  "latitude": 6.5244,
  "longitude": 3.3792,
  "rainfall_mm_24h": 85,
  "river_level_m": 3.2
}
```

Response body (live example)

```
{
  "location_name": "Cairo, Egypt",
  "country": "Egypt",
  "latitude": 30.0444,
  "longitude": 31.2357,
  "rainfall_mm_24h": 35.0,
  "river_level_m": 2.0,
  "risk_level": "medium",
  "risk_score": 0.42,
  "alert_message_en": "Flood risk is MEDIUM in Cairo. Stay alert and monitor local updates.",
  "alert_message_local": "توَعَّ الحذر وتابع التحديثات المحلية Cairo. خطر الفيضانات متوسط في",
  "local_language": "Arabic",
  "timestamp": "2026-08-10T19:03:57Z",
  "risk_score_breakdown": {
    "rainfall_mm_24h": 35.0,
    "river_level_m": 2.0,
    "normalized_rainfall": 0.35,
    "normalized_river_level": 0.5,
    "rainfall_cap_mm": 100.0,
    "river_level_cap_m": 4.0,
    "high_threshold": 0.7,
    "medium_threshold": 0.4,
    "risk_level": "medium",
    "risk_score": 0.42
  },
  "ml_risk_level": "medium",
  "ml_risk_score": 0.43
}
```

Field reference

FIELD	TYPE	MEANING
location_name	string	Echoed from the request, "City, Country"
country NEW	string	Parsed from location_name . Empty string if it has no comma.
latitude / longitude NEW	float	Echoed from the request.

rainfall_mm_24h	NEW	float	Echoed from the request. Same value also inside risk_score_breakdown .
river_level_m	NEW	float	Echoed from the request. Same value also inside risk_score_breakdown .
risk_level	PRIMARY	string	"low" "medium" "high" — the rules-based score. Use this one for alerting logic.
risk_score	PRIMARY	float	0.0–1.0, rules-based.
alert_message_en		string	Ready-to-send alert, English.
alert_message_local		string	Same alert, translated. Right-to-left when local_language is "Arabic" — render with dir="rtl" , this field has no directionality markers itself.
local_language		string	"English" "Swahili" "Arabic" "Somali"
timestamp		string	ISO 8601, UTC.
risk_score_breakdown		object	Explainability data — the normalized inputs and thresholds behind risk_score . Same fields as shown in the example above.
ml_risk_level / ml_risk_score		string / float	A separately trained ML model's second opinion, same value ranges as the rules-based fields. Will often differ slightly — that's expected, not a bug. Not the primary score for alerting.

All 9 monitored sample cities with current risk, computed live. Real, not stubbed.

Response body (live example, 2 of 9 entries shown)

```
[
  {
    "location_name": "Lagos, Nigeria",
    "country": "Nigeria",
    "latitude": 6.5244,
    "longitude": 3.3792,
    "rainfall_mm_24h": 85,
    "river_level_m": 3.2,
    "risk_level": "high",
    "alert_message_en": "Flood risk is HIGH in Lagos. Move to higher ground and avoid riverbanks.",
    "alert_message_local": "Flood risk is HIGH in Lagos. Move to higher ground and avoid
riverbanks.",
    "local_language": "English",
    "population_estimate": 15000000,
    "risk_score_breakdown": { "...": "same shape as above" },
    "ml_risk_level": "high",
    "ml_risk_score": 0.84
  },
  {
    "location_name": "Nairobi, Kenya",
    "country": "Kenya",
    "latitude": -1.2921,
    "longitude": 36.8219,
    "rainfall_mm_24h": 12,
    "river_level_m": 1.1,
    "risk_level": "low",
    "alert_message_en": "Flood risk is LOW in Nairobi. No action needed at this time.",
    "alert_message_local": "Hatari ya mafuriko ni NDOGO katika Nairobi. Hakuna hatua inayohitajika
kwa sasa.",
    "local_language": "Swahili",
    "population_estimate": 4400000,
    "risk_score_breakdown": { "...": "same shape as above" },
    "ml_risk_level": "low",
    "ml_risk_score": 0.2
  }
  // ... 7 more, same shape – 9 total
]
```

Field reference

FIELD	TYPE	MEANING
location_name , latitude , longitude , risk_level	—	Unchanged since the skeleton — safe to already be relying on these.
country NEW	string	Parsed from location_name .
rainfall_mm_24h / river_level_m NEW	float	This region's sample sensor input.

<code>alert_message_en</code> / <code>alert_message_local</code> / <code>local_language</code> NEW	string	Computed the same way as <code>/api/risk-check</code> . This is the only place in this endpoint an alert message appears — see the note below about <code>/api/alerts</code> .
<code>population_estimate</code>	int	Rough general city population. Not a flood-exposure number — don't present as "people at risk."
<code>risk_score_breakdown</code>	object	Same shape as in <code>/api/risk-check</code> .
<code>ml_risk_level</code> / <code>ml_risk_score</code>	string / float	Same meaning as in <code>/api/risk-check</code> .

DON'T CONFUSE THIS WITH GET /API/ALERTS

`/api/alerts` is a separate, simulated, 5-entry "alert history" stub — it is NOT one entry per region and is not computed live. If you want an alert message per region for a map/list view, use the `alert_message_*` fields on *this* endpoint, not `/api/alerts` .

GET /api/alerts

Simulated "recently sent" alert history, for a history/feed view. **Intentionally stubbed** — no real SMS/USSD/WhatsApp gateway this week. Returns a fixed 5-entry list, not computed live, not one-per-region.

Response body (live example, 1 of 5 entries shown)

```
[
  {
    "location_name": "Lagos, Nigeria",
    "risk_level": "high",
    "message_sent": "Flood risk is HIGH in Lagos. Move to higher ground and avoid riverbanks.",
    "channel": "SMS (simulated)",
    "timestamp": "2026-08-07T09:15:00Z"
  }
  // ... 4 more — Kampala/USSD, Dar es Salaam/SMS, Accra/WhatsApp, Cairo/SMS
]
```

Field reference

FIELD	TYPE	MEANING
location_name	string	
risk_level	string	"low" "medium" "high"
message_sent	string	The exact text that "was sent."
channel	string	"SMS (simulated)" "USSD (simulated)" "WhatsApp (simulated)" — the only place channel type appears.
timestamp	string	ISO 8601, UTC.

LIVING SOURCE OF TRUTH

This PDF is a snapshot as of 2026-08-10. The canonical, always-current version is `docs/api-contract.md` in the repo — if the schema changes again, check there (or ask for this PDF to be regenerated) rather than trusting this file indefinitely.